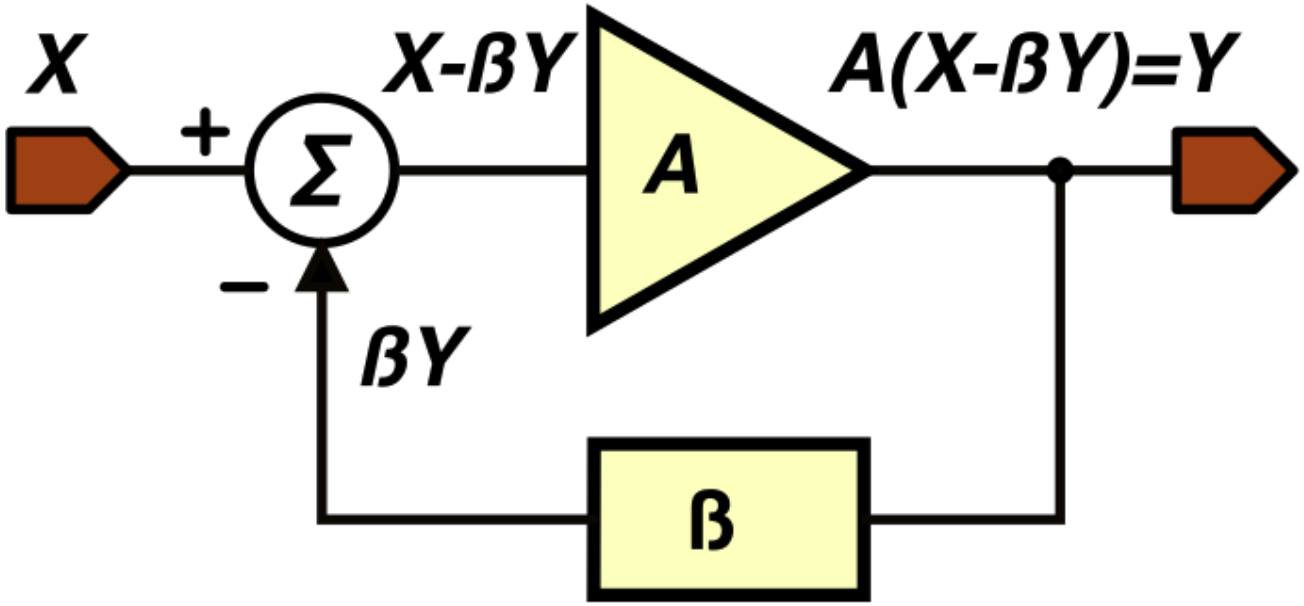
Project Brief – 2

Chen, Junzhe (JC)

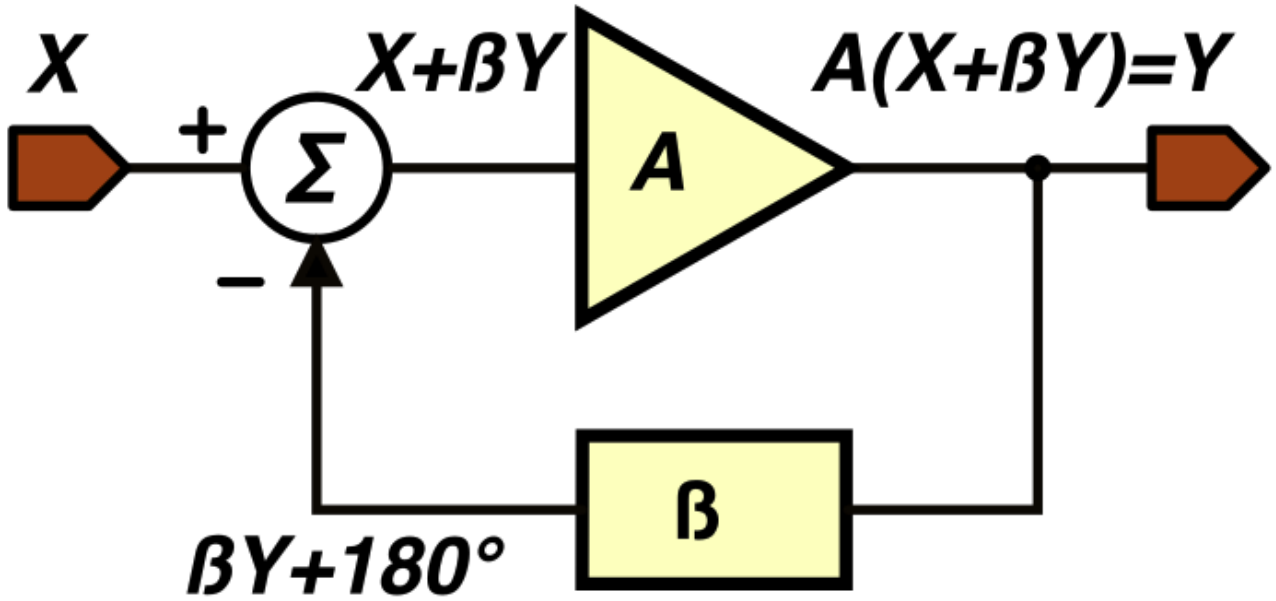
09/Nov/2026, Bristol, UK

Negative feedback



$$\frac{Y}{X} = \frac{forward\ gain}{1 - loop\ gain} = \frac{A}{1 + A\beta}$$

Negative feedback + phase shift

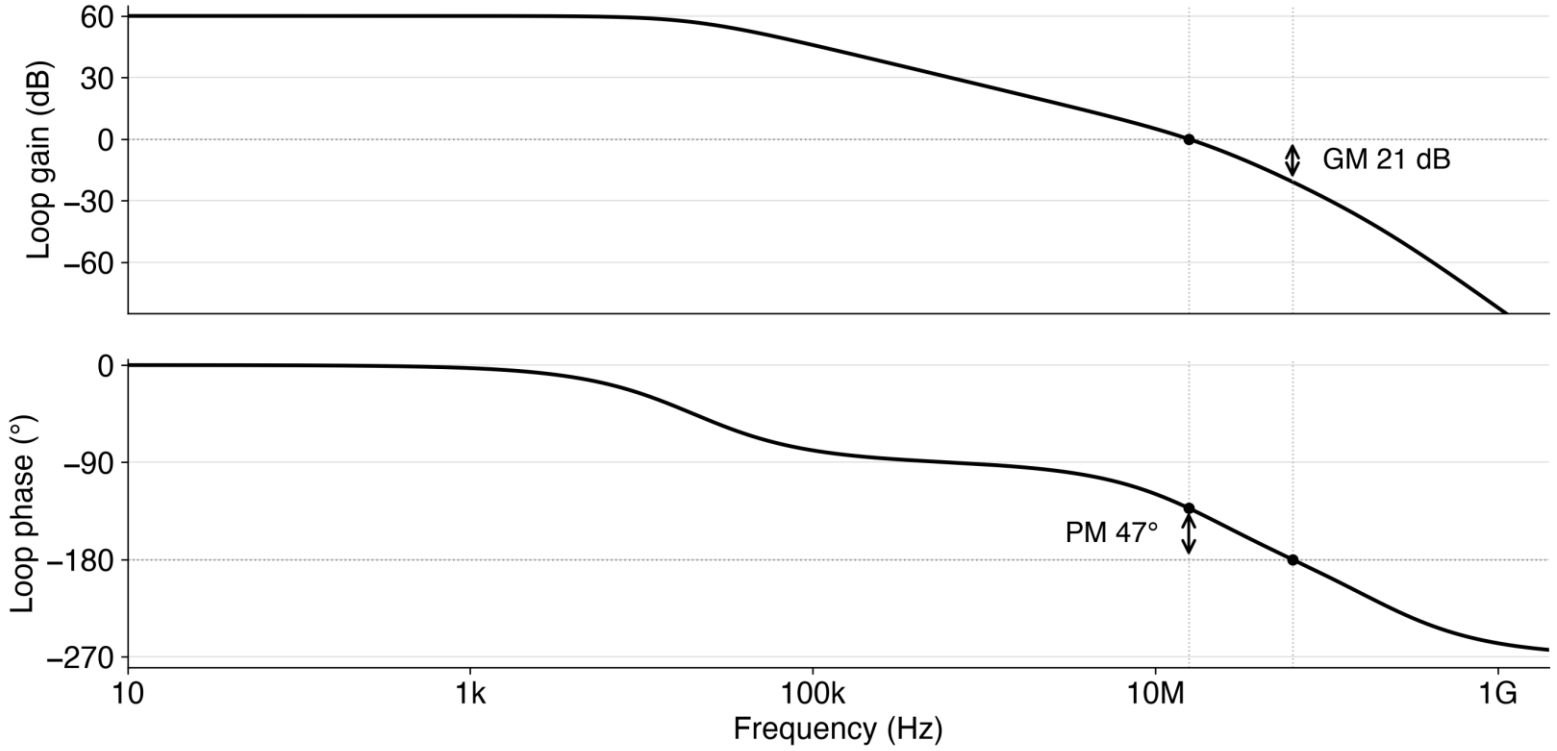


$$\frac{Y}{X} = \frac{forward\ gain}{1 - loop\ gain} = \frac{A}{1 - A\beta}$$

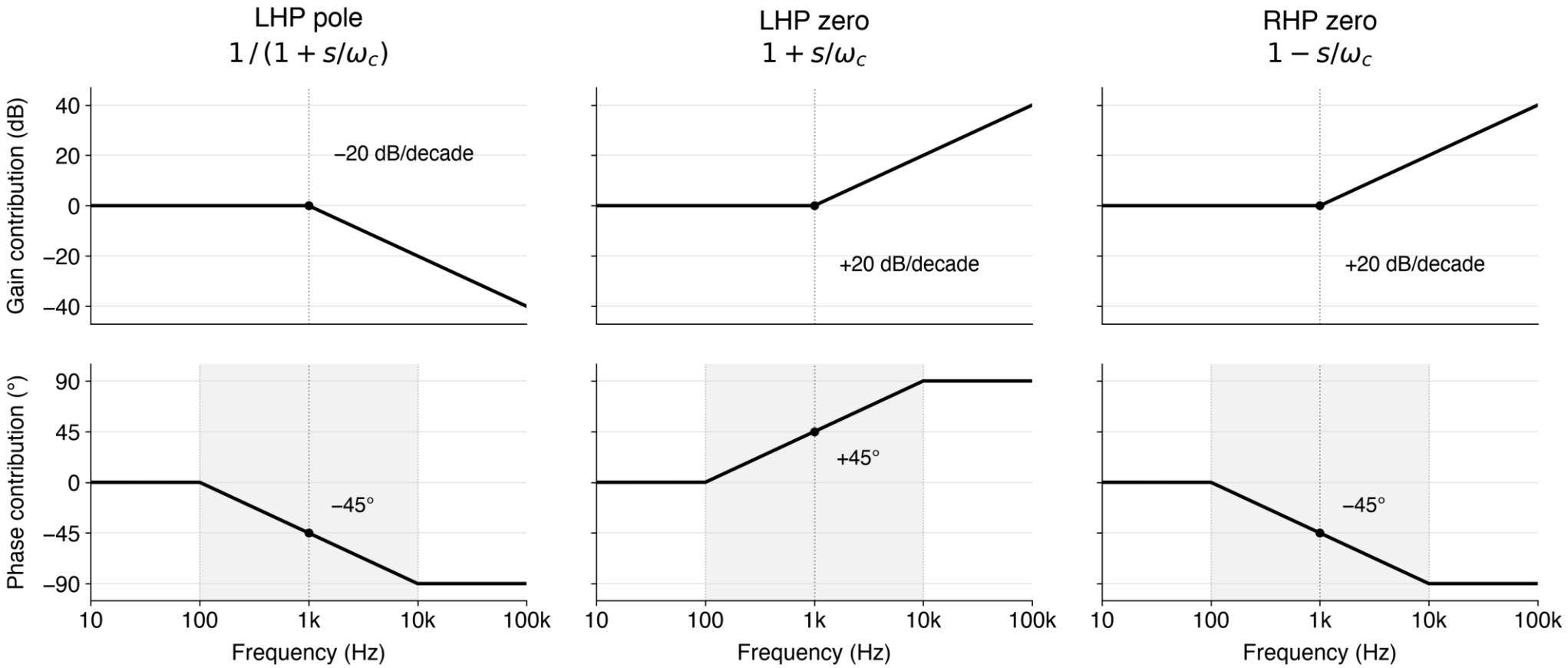
- Dangerous if $A\beta = 1$

Stability I

- **Phase margin:** phase away from -180º when gain is zero
- **Gain margin:** gain beyond zero when phase is -180º



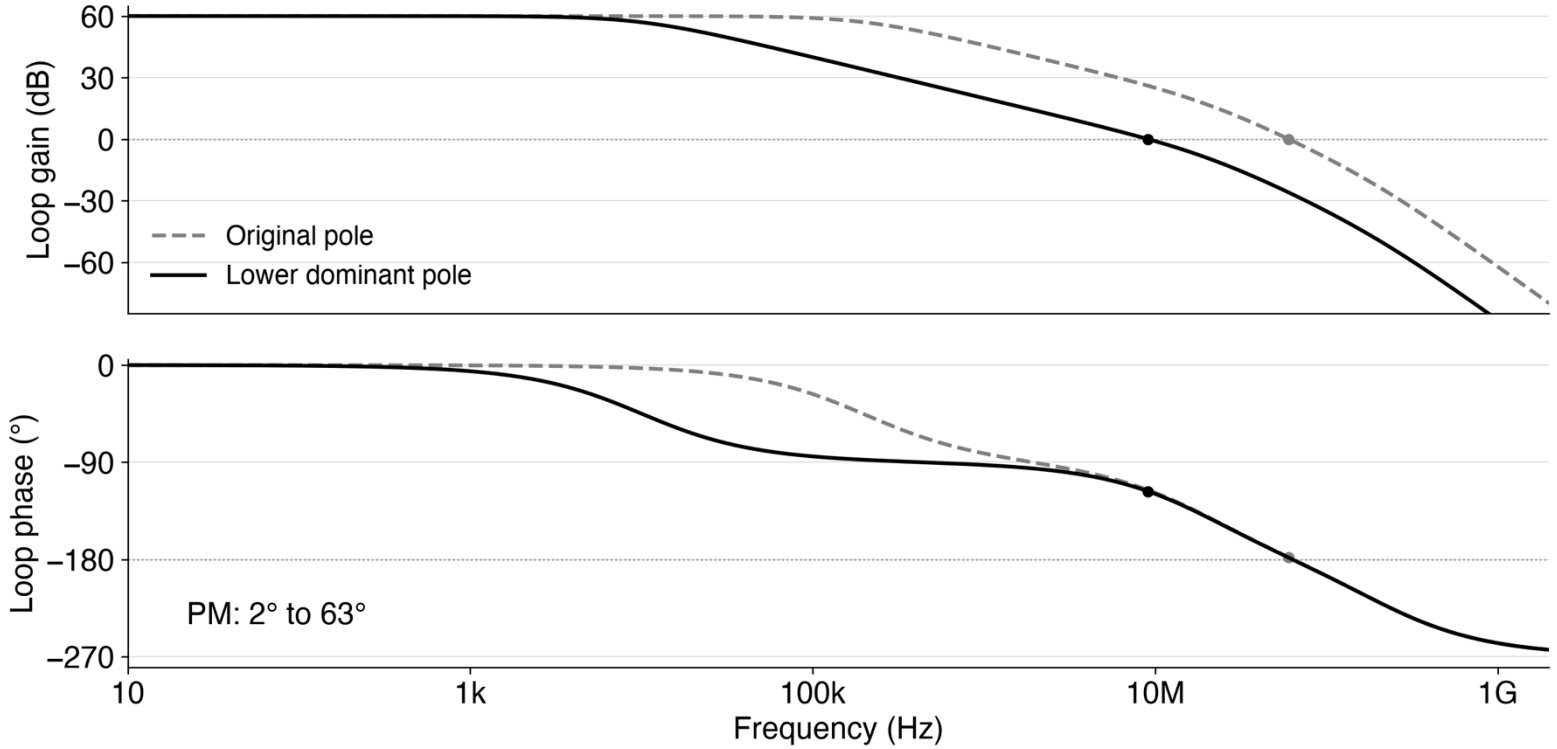
Stability II



Straight-line approximation $f_c = 1$ kHz Phase transition: $0.1f_c$ to $10f_c$

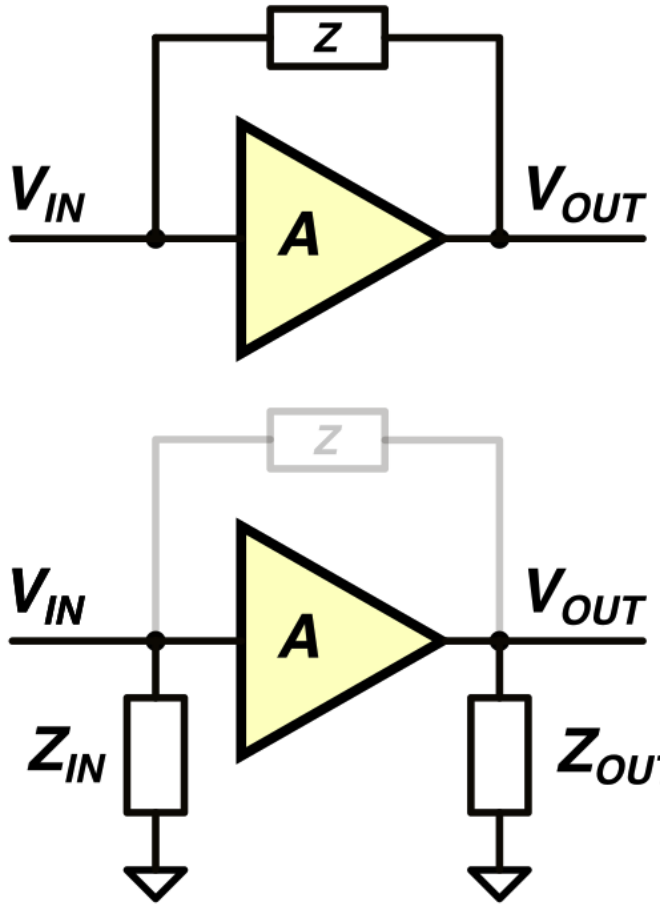
Stability III

- **Dominant pole:** pole @ lowest frequency (2 decades before next)



Impedance over gain

- Break impedance across into
  - Impedance @ input
  - Impedance @ output



By Gain equation

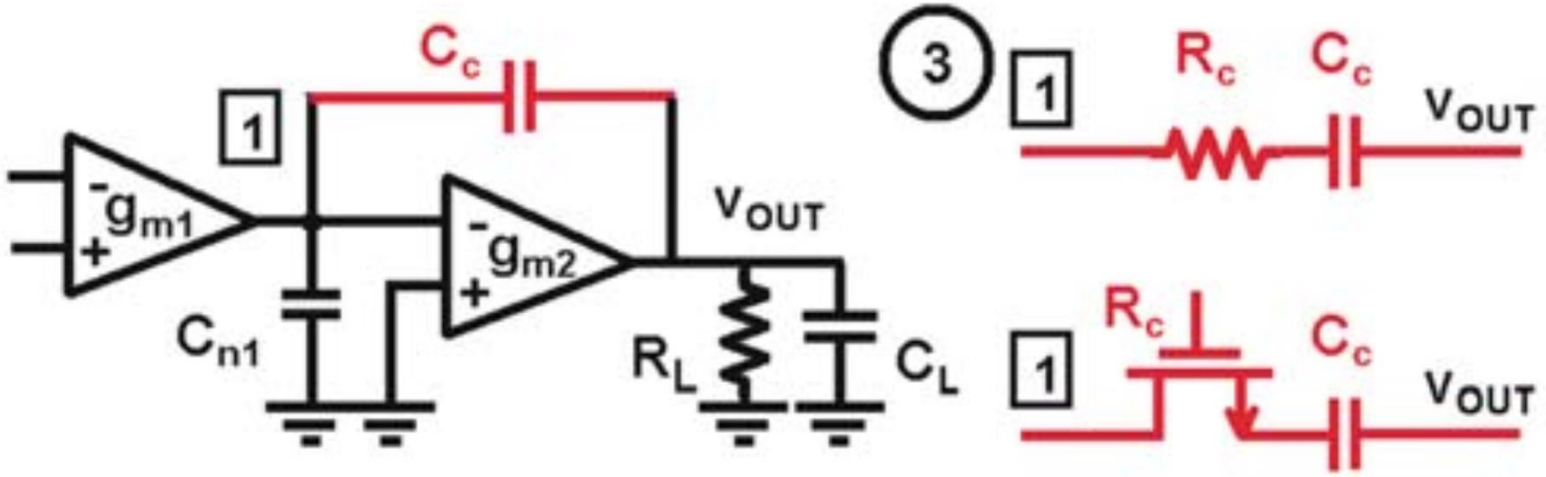
$$V_{OUT} = A \cdot V_{IN}, \qquad V_{IN} = V_{OUT}/A$$

By Ohm's Law

$$Z = V/I \quad \Rightarrow \quad Z_{IN} = \frac{Z}{1-A}, \qquad Z_{OUT} = \frac{Z}{1-1/A}$$

Stability IV

- RHP zero (capacitor feedforward) cancelled by resistor



$$f_z = \frac{1}{2\pi C_c (1/g_{m2} - R_c)}$$

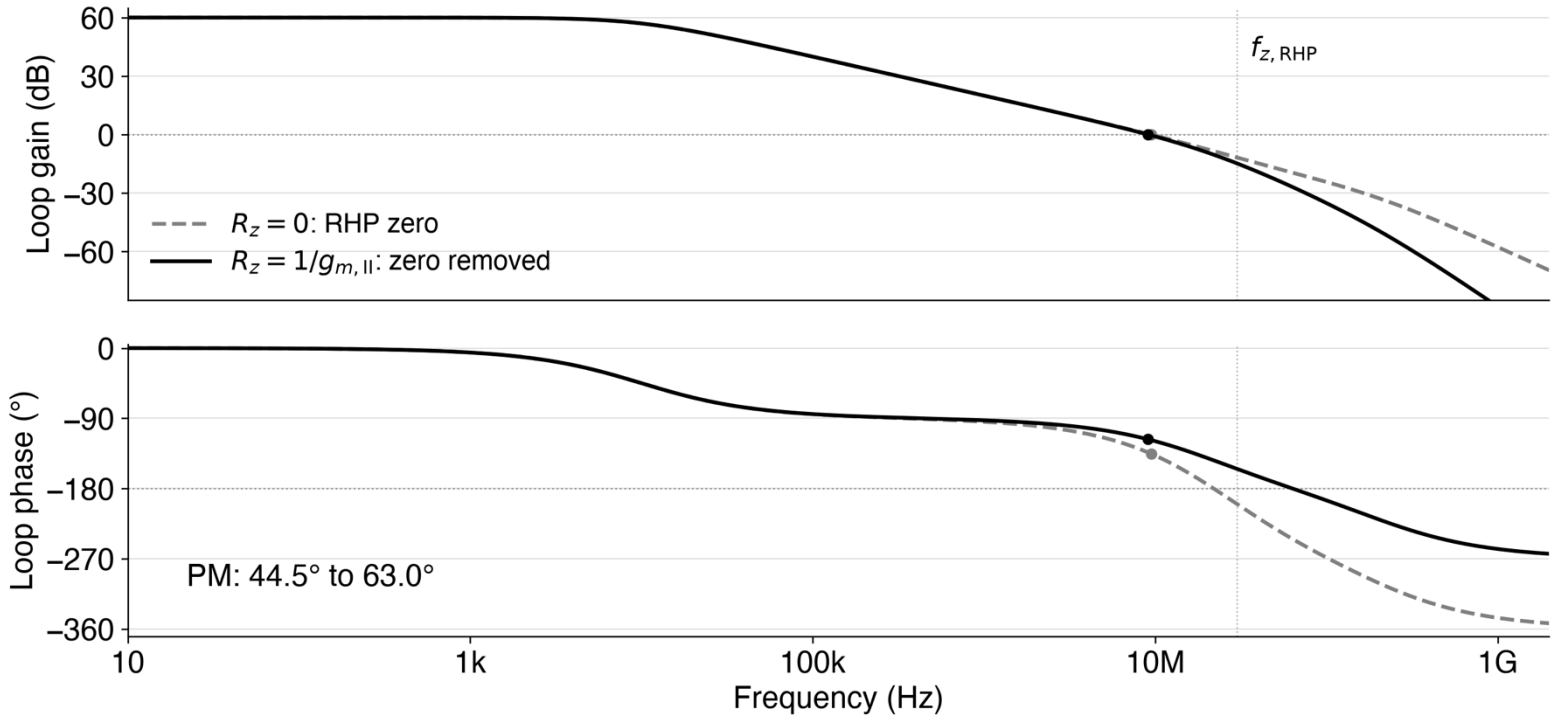
$R_c = 1/g_{m2}$ No zero

$R_c > 1/g_{m2}$ Negative zero

Sasnen slides 0546 (p. 173)

Stability V

- RHP zero (capacitor feedforward) cancelled by resistor



Illustrative model: same dominant pole; other poles fixed; resistor pole neglected.

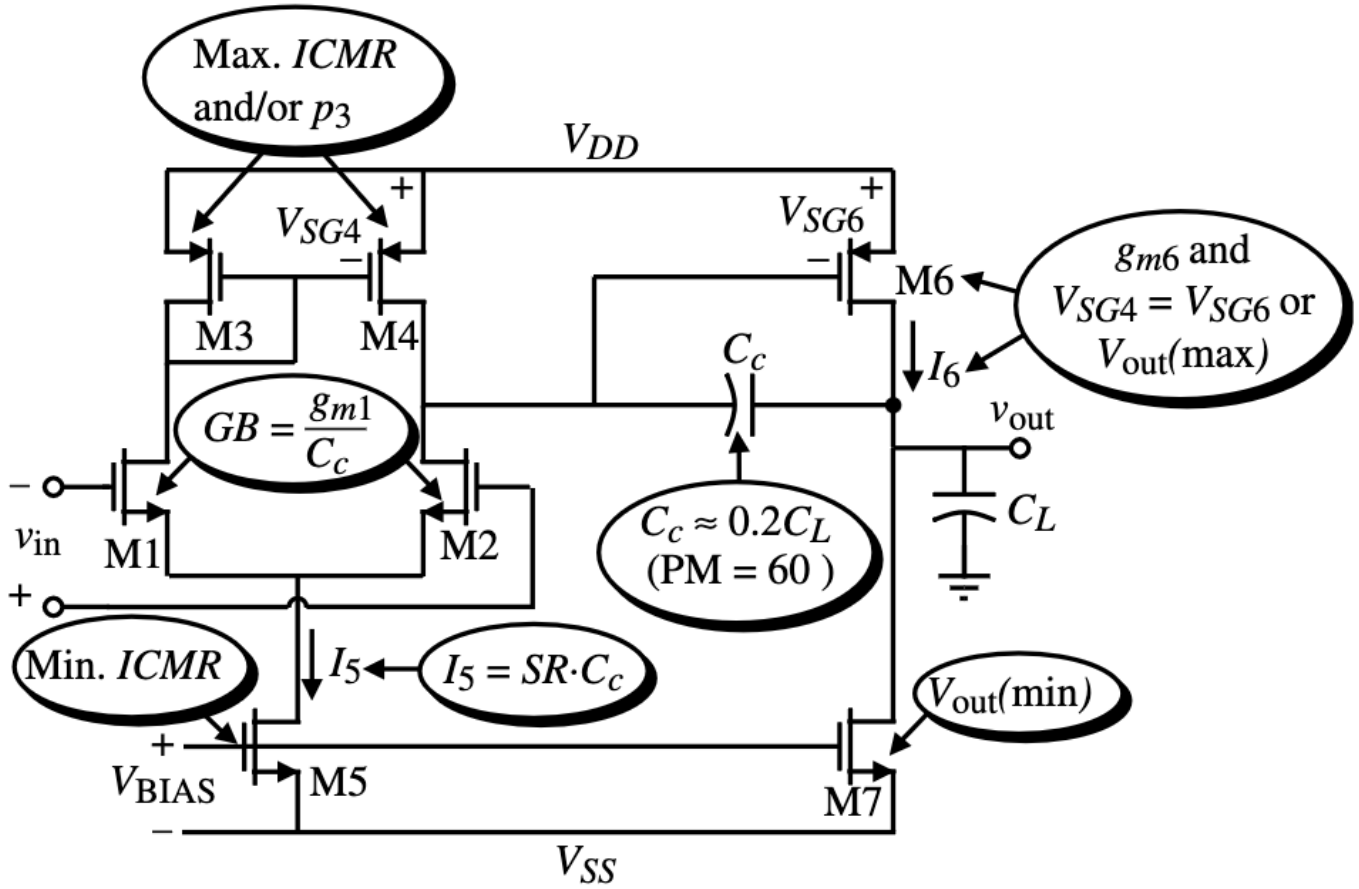
OpAmp selection

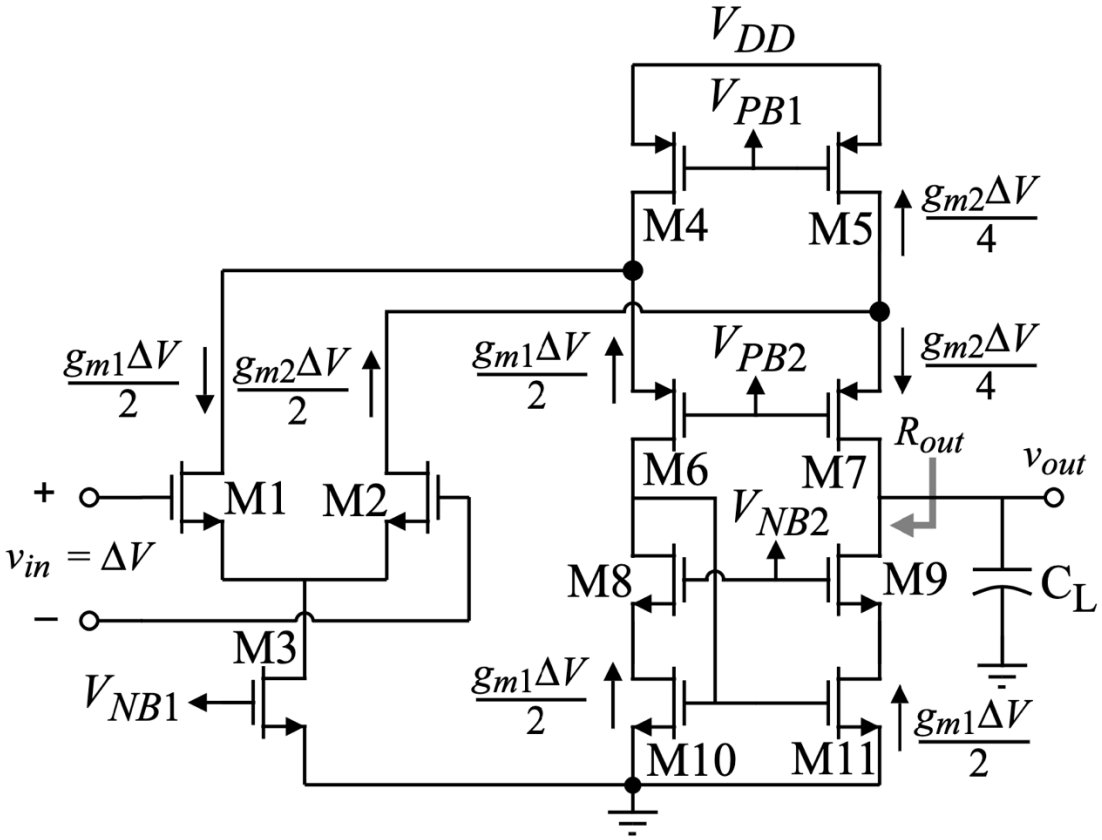
- **PMOS or NMOS input:** voltage headroom
- **5-T OTA:** Low gain, low-power
- **Two-stage:** High Gain & headroom, harder to compensate
- **Folded-cascode:** High Gain, low output headroom

Read the Razavi’s article (GitHub resource)

Hand Calculation example

- Philip Allen p.290, p.318





Develop Intuition

- Philip Allen p.291

Table 6.3-1 Dependence of the Performance of Fig. 6.3-1 on dc Current, W/L Ratios, and the Compensating Capacitor

	Drain Current		M1 and M2		M3 and M4		Inverter	Inverter Load		Compensation Capacitor
	I_5	I_7	W/L	L	W	L	W_6/L_6	W_7	L_7	C_c
Increase dc Gain	$(\downarrow)^{1/2}$	$(\downarrow)^{1/2}$	$(\uparrow)^{1/2}$	$\uparrow$		$\uparrow$	$(\uparrow)^{1/2}$		$\uparrow$	
Increase GB	$(\uparrow)^{1/2}$		$(\uparrow)^{1/2}$							$\downarrow$
Increase RHP Zero		$(\uparrow)^{1/2}$					$(\uparrow)^{1/2}$			$\downarrow$
Increase Slew Rate	$\uparrow$									$\downarrow$
Increase C_L										$\uparrow$